Same Data, Two Analysts: Comparing Generative AI Tools with a Scripted Pipeline on Midjourney Prompts

COMP 9200: Applied Research Methods
British Columbia Institute of Technology
Burnaby, Canada

Christopher Elliott celliott64@my.bcit.ca
Mitchell Winter mwinter18@my.bcit.ca
Boris Kim skim743@my.bcit.ca

***Abstract*—This study analyses 246,381 rows of user prompts from the Midjourney Discord server with two generative AI tools (GitHub Copilot and Claude Code), each given only the data and a fixed sequence of questions, and with a scripted Python pipeline written by the authors. 51.2% of rows are not prompts, 73.5% are exact duplicates, and 83.2% of test rows also appear in the training split. For the 57,670 clean prompts, the paper reports prompt structure, artist references, quality-booster phrases and topics. The AI tools found the same data problems, used effect sizes and avoided causal claims, but differed from the scripted pipeline wherever a definition had to be chosen: what counts as a prompt, how length is counted, and which artists and phrases are searched for. On one hypothesis this turned a negligible effect into a 21-point difference. An AI analyst is one more analyst with its own hidden choices, and its definitions need the same scrutiny as its numbers.**

***Index Terms*—generative AI, data analysis, reproducibility, text-to-image prompts, data quality, topic modelling**

I. Introduction

Generative AI tools can now load a data file, write and run analysis code, and summarise the results in plain language. This makes them attractive as research assistants, but it raises questions about rigor, transparency and reproducibility: the analyst may not see the decisions the tool made, and the same request can give different answers on different days.

This study gives the same public dataset and the same research questions to two generative AI tools and to a scripted pipeline, and compares the results. The dataset is a collection of text-to-image prompts from the Midjourney Discord server [1]. It suits the comparison because it has a single free-text column, so every variable has to be derived by rules the analyst chooses.

The research questions are:

- **RQ1:** What does a typical prompt look like (length, parameters, aspect ratios)?
- **RQ2:** How often do prompts name an artist or use generic "quality booster" phrases?
- **RQ3:** What themes do prompts cover, and how stable are the topics found?
- **RQ4:** Are parameter use and prompt length related, and are artist references and booster phrases related?

For each question, the AI tools and the scripted pipeline are compared on findings, rigor, transparency, bias and reproducibility.

II. Related Work

Text-to-image prompts. Liu and Chilton derived design guidelines for prompting text-to-image models [2], and Oppenlaender proposed a taxonomy of prompt modifiers such as style and quality boosters [3]. Large prompt collections such as DiffusionDB [4] and prompt-log studies [5] describe how users write prompts in practice. Both describe prompts as blocks of subject and style terms, with the same style terms and artist names (e.g. "highly detailed", "artstation") recurring across many prompts, which RQ1 and RQ2 measure directly.

AI as data analyst. Cheng et al. evaluated GPT-4 as a data analyst [6]: it matched or beat junior analysts on most criteria but trailed senior analysts, and it stated calculation errors in a confident tone, so they came to light only when the work was checked.

Analytic flexibility and data quality. Silberzahn et al. showed that independent analysts given the same data reach different conclusions [7]; an AI tool is one more analyst whose choices may be hidden. Duplicates in text corpora inflate evaluation results [8], and datasheets [9] were proposed to document how a dataset was collected and what it should be used for.

III. Methodology

A. Dataset

`succinctly/midjourney-prompts` [1] contains 246,381 rows with a single `text` column, split into train (221,743), validation (12,318) and test (12,320). It was derived from a Kaggle scrape of the public Midjourney Discord server [10], keeping only the user's text. It has no user identifiers, timestamps or images, and is released under Apache-2.0. The Parquet files were downloaded on 6 Nov. 2026 and their SHA-256 hashes recorded (Appendix B).

B. Preprocessing

All rules are defined in one file (`prompt_utils.py`) so they can be quoted and checked.

TABLE I
ROW TYPES IN THE RAW DATA (ALL SPLITS).

Row type	Rows	%
prompt	120,357	48.8
short fragment	59,726	24.2
aspect ratio only	34,571	14.0
url or params only	14,832	6.0
number only	13,922	5.7
empty	1,597	0.6
symbols only	1,376	0.6

TABLE II
FROM RAW ROWS TO CLEAN PROMPTS.

Step	Rows
Raw rows (all splits)	246,381
excluded: not a prompt	126,024
Prompt rows	120,357
excluded: duplicate	62,687
Clean prompts	57,670

- *Parameters* are tokens of the form `--name value`. Because phones and Discord often turn `--` into an en or em dash, all three are accepted. Names are mapped to canonical forms (e.g. `--ar` and `--aspect` to *aspect*).
- *Content words* are the alphabetic tokens left after removing parameters, URLs and `::` weights.
- *Row type.* A row is a *prompt* if it has at least three content words; otherwise it is classed as empty, aspect ratio only, number only, URL or parameters only, symbols only, or a short fragment (Table I).
- *Duplicates* are found after lower-casing and collapsing whitespace. The first copy is kept.

Because the provided splits overlap (Section IV-A), the splits are pooled and repeated prompts removed, leaving 57,670 clean prompts (Table II). No model is trained, so the analysis needs no new split; a model built on these data would need the deduplicated prompts re-split, with near-duplicates kept in the same split.

C. AI Tool Setup

Run A (8 Nov. 2026) used the GitHub Copilot agent in Visual Studio Code; the chat export records the model only as `gpt-6.1-terra`, with no published version. *Run B* (11 Nov. 2026) used the Claude Code extension with `claude-opus-5-5` at medium effort. Run B is a second analyst, not a repeat of run A, and since a Claude model also helped write the scripted pipeline, it is the less independent AI arm. Each run started a fresh session with memory, custom instructions and web search off (by the operator's account; the exports do not record this). The tool worked in a local folder holding one file, `midjourney_prompts_all.csv` (columns `split` and `text`, 246,381 rows), ran its own Python code, and was not given the pipeline's rules or results. It received seven fixed prompts (P0–P6): an open-ended request, one per research question, and a request for the complete code (Appendix A), with no hints in between. Each response was saved unedited and scored against the pipeline's results on findings, rigor, transparency, bias and reproducibility.

D. Scripted Pipeline

The pipeline is a Python script (`pipeline.py`) using pandas, SciPy and scikit-learn [11], with pinned library versions and a fixed random seed. It checks the input files' hashes, applies the preprocessing above, and writes every reported number, table and figure, so the paper cannot drift from the code.

RQ1 reports medians and interquartile ranges of content words, the share of prompts using each parameter, and aspect-ratio counts.

RQ2 detects artist references with two rules, to show how much the answer depends on the rule: a *strict* rule ("by" or "in the style of" followed by capitalised words) and a *lenient* rule (the same cue words in any case, split on "and" and commas). Quality boosters are matched against a fixed list of 27 phrases taken from common prompt advice (e.g. "trending on artstation", "8k", "octane render").

RQ3 fits a TF-IDF representation (minimum document frequency 20, English stop words removed) and non-negative matrix factorisation (NMF) [12] with $k = 10$ topics. To test stability, the model is refitted on five random 80% subsamples, each started from the same SVD-based initialisation as the reference model (NNDSVDar); topics are matched to the reference run, and the Jaccard overlap J of their top ten words is reported.

RQ4 tests H1 (prompts with any parameter are longer) with a Mann–Whitney U test and rank-biserial correlation, and H2 (artist references, under the lenient rule, are associated with booster phrases) with a χ^2 test and Cramér's V. With $n = 57,670$, almost any difference is significant, so effect sizes, not p-values, are interpreted.

IV. RESULTS

Table III sets the headline numbers side by side. The subsections give the scripted pipeline's results and then the AI tools'.

A. Data Quality

Only 120,357 of 246,381 rows (48.8%) are prompts. The rest are bare aspect ratios such as `16:9`, numbers, lone parameters such as `--uplight`, and one- or two-word fragments (Table I). The median raw row is 15 characters long. 180,983 rows (73.5%) are exact duplicates of an earlier row, and 83.2% of test rows (68.4% of test rows that are real prompts) also appear in train. A model trained and evaluated on these splits would be tested largely on data it had already seen [8].

B. RQ1: Prompt Structure

Clean prompts have a median of 12 content words (IQR 7–21), with a long right tail (Fig. 1). 47.6% use at least one parameter; aspect ratio is by far the most common (36.1%), led by 16:9 and 9:16 (Fig. 2). 8.3% use `::` weights. Most prompts are comma-separated lists of descriptors rather than sentences.

TABLE III
HEADLINE RESULTS: AI TOOLS (RUNS A AND B) VS. THE SCRIPTED PIPELINE.

Question	Measure	Run A (Copilot)	Run B (Claude)	Scripted pipeline
Data quality	Exact duplicate rows	180,983 (73.5%)	180,983 (73.5%)	180,983 (73.5%)
	Test rows also in train	83.1%	83.1%	83.2%
	Rows analysed after cleaning	64,907	59,733	57,670
RQ1	Median length (IQR)	12 (6–21)[a]	12 (7–20)	12 (7–21)
	Any parameter	42.2%	46.2%	47.6%
	`--ar`/`--aspect`	31.8%	35.3%	36.1%
RQ2	Names an artist	13.5% (50 names)	21.3% (168 names)	14.4% strict, 24.3% lenient
	Booster phrase	21.8% (6 phrases)	33.1% (16 phrases)	36.2% (27 phrases)
RQ3	Topic stability	cosine 0.88, 0.76	$J = 0.75$ (subsamples)	$J = 0.81$ (subsamples)
RQ4	H1 effect size	$r_{rb} = 0.60$	$r_{rb} = 0.41$	$r_{rb} = 0.40$
	H2 effect size	OR $= 2.84$ ($V = 0.17$)[b]	OR $= 2.17$ ($V = 0.16$)[c]	$V = 0.08$

[a]Whitespace items, counting parameter flags and values as words; run B and the scripted pipeline count content words only. [b]The tool reported the odds ratio; V is computed from the tool's χ^2 and n. [c]Falls to OR ≈ 1.0 when prompt length is controlled.

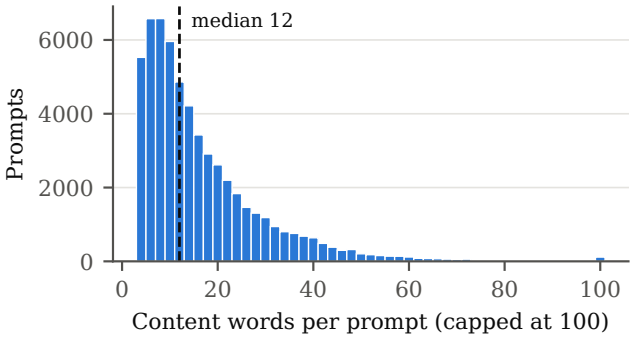


Fig. 1. Content words per clean prompt.

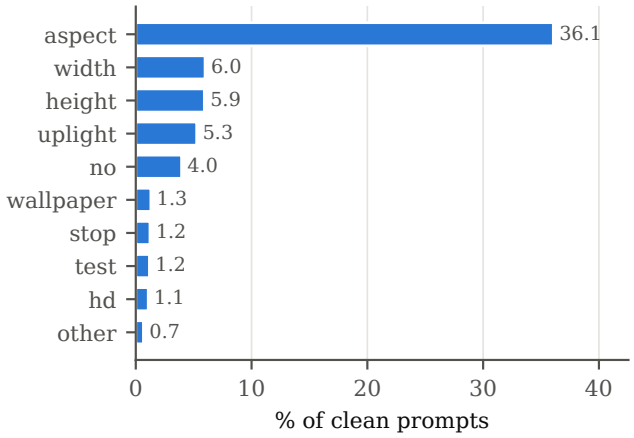


Fig. 2. Share of clean prompts using each parameter.

C. RQ2: Artists and Quality Boosters

The share of prompts naming an artist is 14.4% under the strict rule and 24.3% under the lenient rule: the rule alone changes the answer by a factor of 1.7. The most-cited names are concept and fantasy illustrators such as Craig Mullins and Ashley Wood (Fig. 3). 36.2% of prompts use at least one booster phrase, most often "cinematic", "8k" and "artstation".

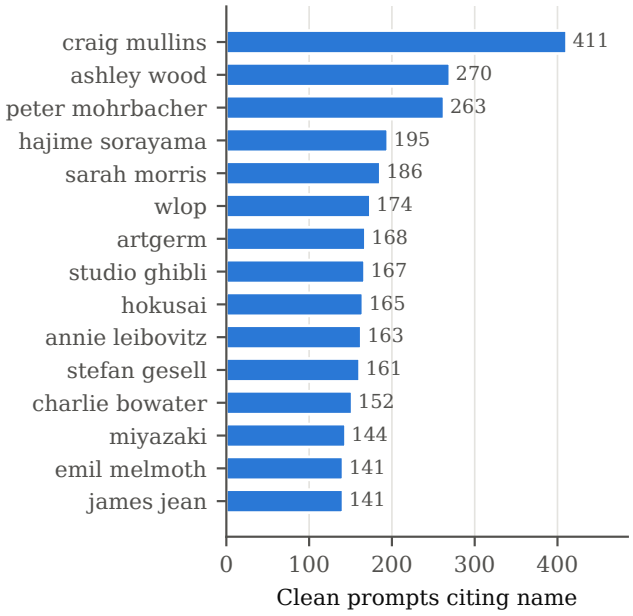


Fig. 3. Most-cited artist or studio names (lenient rule).

D. RQ3: Topics

Table IV lists the ten NMF topics. They separate rendering style (octane/unreal render, oil painting, black-and-white ink, hyper-realistic photo, vector art) more than subject matter, which is consistent with booster phrases dominating word frequencies. Mean stability across subsamples is $J = 0.81$; most topics are stable ($J > 0.7$), but the least stable topic, a mix of artists' first names, is not ($J = 0.52$).

E. RQ4: Hypothesis Tests

Prompts using a parameter are longer (median 17 vs. 9 content words; Mann–Whitney $p < .001$, $r_{rb} = 0.40$, a

TABLE IV
NMF TOPICS: SHARE OF PROMPTS, STABILITY J, TOP WORDS.

#	%	J	Top words
1	7.8	0.80	render, octane, unreal, engine, high, artstation
2	10.8	0.73	style, ghibli, anime, poster, design, city
3	1.0	0.52	artwork, van, john, peter, mike, robert
4	8.6	0.93	art, concept, fantasy, circa, pulp, magazine
5	9.4	0.84	painting, oil, matte, abstract, space, beautiful
6	19.2	0.93	black, white, background, red, ink, photo
7	12.5	0.85	portrait, beautiful, artstation, character, face, girl
8	9.9	0.79	realistic, hyper, photo, ultra, old, epic
9	16.6	0.72	detailed, cinematic, highly, lighting, intricate, dark
10	3.2	1.00	vector, psychedelic, flat, rule, symmetrical, thirds

medium effect). Artist references and booster phrases are associated ($\chi^2(1) = 347.8$, $p < .001$), but the effect is negligible ($V = 0.08$): 42.8% of prompts naming an artist use a booster, against 34.1% of those that do not.

F. AI Tool Results

Run A (Copilot). Unprompted (P0), the tool led with data quality: 73.5% of rows repeat an earlier row and 83.1% of test rows occur in train, so "the existing splits [are] unreliable for measuring generalization". Its duplicate count matched the pipeline's exactly (180,983). In P1 it flagged a row as "not a real prompt" only if the whole row was an aspect ratio, number, parameters or URL (25.5% of rows), noting that 52,970 rows of at most one word were left in; the pipeline's three-word rule excludes 51.2%. This left 64,907 distinct rows, 13% more than the pipeline's. Its median length matches the pipeline's (12), but it counted whitespace items including parameter flags, and 7.9% of its rows were a single item; these short rows rarely carry parameters, which lowers its parameter shares. The top aspect ratios agree (16:9, 9:16, 2:3). For artists (P3) it mined names after cues such as "in the style of" and then searched a fixed list of 50; the mining output was not kept, so the list cannot be traced to a rule. It found 13.5%, just below the pipeline's strict rule (14.4%), with the same top name (Craig Mullins). Its six booster phrases (three from the prompt, three its own) found 21.8%. For topics (P4) it chose the pipeline's method and settings (TF-IDF, NMF, $k = 10$) and found nearly the same topics, down to a 0.9% topic of artists' first names; two reruns gave matched cosine similarity 0.88 and 0.76, which it called "not highly stable". For H1 (P5) it reported $r_{rb} = 0.60$, and noted that its length measure counts the parameters themselves (0.49 without them). For H2 it reported an odds ratio of 2.84 (39.9% vs. 18.9% booster use) and $p \approx 1.7 \times 10^{-3677}$, a spuriously precise figure. It called both results "associations, not causal findings". Its P6 code was "a consolidated, verified reproduction, not a verbatim transcript"; run with the pipeline's library versions, it reproduced every number in P1–P5.

Run B (Claude). Run B matched run A's exact counts but chose definitions much closer to the pipeline's, and so did its numbers (Table III). It treated a row as a fragment only if it was a single word, supported by showing that 96.7% of such rows occur inside longer prompts, which left 59,733 prompts (4% more than the pipeline's). It removed parameters before counting words, so its median and H1 effect ($r_{rb} = 0.41$) almost match the pipeline's. For artists it combined a wording rule (17.2%) with 168 names mined from the data (21.3%), checking ambiguous names by hand (e.g. excluding "Möbius strip"). For topics it chose k by coherence and stability, fitted a second model without style words, and compared NMF with LDA (top-word overlap 0.17–0.23), concluding that topic shares "can move by tens of points between runs". On H2 it found the same raw association as run A (OR $= 2.17$) but then controlled for prompt length, since artist prompts are longer: the odds ratio fell to 0.95 (logistic regression) and 1.01 (Mantel–Haenszel). It also claimed the data show a content filter, which the data alone cannot establish. The tool reports that its rewritten P6 code reproduced every number; a separate re-run confirmed this.

V. Discussion

A. Interpretation

The two analyses agree wherever the question has one obvious answer: exact duplicates, train–test leakage, the most common aspect ratios and artists, and, given the same model settings, the topics. They differ wherever the analyst must define a variable, and every difference in Table III traces to one of four decisions: (1) the *non-prompt rule*, where run A kept short fragments that lower its parameter shares; (2) the *length unit*, where counting parameter tokens as words inflates the H1 effect from 0.40 to 0.60; (3) the *artist rule*; and (4) the *booster list*. Decisions (3) and (4) together change the H2 conclusion. The scripted pipeline finds the association negligible ($V = 0.08$, a 9-point gap); run A's figures (OR 2.84, a 21-point gap) would usually be read as a real, small-to-moderate effect; run B adds a third reading, that the association almost vanishes once prompt length is controlled. None of these is wrong; each answers a differently defined question. This is the "many analysts" result [7] with AI tools as two of the analysts.

On rigor, both tools did better than expected: they found the data problems unasked, reported effect sizes, ran sensitivity checks and stated the limits of their rules. Their weaknesses lay in what they left unseen: exploratory steps that shaped the artist lists, and final code rewritten after the fact. Run A also made one numerical error, which it caught itself in P6.

B. Validity

Construct validity. "Names an artist" and "uses a booster" depend on the rule. The strict and lenient rules differ by a factor of 1.7, and the AI tools' lists give further answers.

To estimate how well the rules work, two of the authors each labelled the same random sample of 100 clean prompts without seeing the rules' output, then settled disagreements by discussion. Against the agreed labels, the strict artist rule has precision 1.00 and recall 0.31, the lenient rule 0.68 and 0.47, and the booster rule 0.85 and 0.78. The raters agreed on 82 of 100 prompts for artists (Cohen's $\kappa = 0.60$, moderate agreement) and on 87 for boosters ($\kappa = 0.72$, substantial). The strict rule never flags a prompt without an artist but misses most artist references, and the lenient rule finds more at the cost of false matches. The raters themselves disagreed mostly on bare names with no cue word and on films or comics named as a style (e.g. "Watchmen"), so "names an artist" is not a sharp category even for people. *Internal validity.* RQ4 tests associations only; prompts were not assigned to conditions, and near-duplicate templates break the independence the tests assume, so the p-values are too small. *External validity.* The data come from one Discord community in 2022, mostly in English, using early Midjourney versions. They do not describe prompting today or on other tools.

C. Bias

Sampling. The prompts come from early adopters in one Discord community, and repeated templates dominate: one prompt appears 736 times. Deduplication counts each distinct prompt once, but near-identical variants of one template still form whole topics, as run B noted. *Measurement.* Each rule favours some prompts over others. The strict artist rule needs capitalised names, so it misses names typed in lower case (recall 0.31), and the booster list finds only phrases that prompt advice already names. *AI tools.* The tools anchored on the question wording: P3 named three example phrases, and run A's booster list was those three plus three more. Both reported p-values with spurious precision, and run B presented a guess about a content filter as evidence. *Evaluator.* The comparisons rest on code output rather than judgement, the hand check was done by people, and the authors reviewed the scoring. The pipeline itself is one analyst's set of choices, so each difference measures distance from the pipeline, not from the truth.

D. Ethics

The Discord users who wrote these prompts did not consent to research use. The dataset drops user names, but prompts can still contain personal names or identifying text, so only aggregates are reported and no prompt that names a private person is quoted. Many prompts ask for the style of living artists, which is the centre of an open dispute over consent and copyright; this study counts these names but does not judge the practice. Some prompts may be sexual or violent; they were not filtered, but the examples in this paper avoid them. The Apache-2.0 licence was applied to scraped third-party text, so it may not settle the authors' rights. Run A raised none of these issues, even when asked open-ended questions; run B raised only the copyright point, noting that many named artists are living illustrators.

E. Reproducibility and AI Transparency

Re-running the scripted pipeline rebuilds every number in this paper (Appendix B). The AI arm is harder to repeat. Run A's model is known only by an internal name and may change without notice, and settings such as memory were recorded by the operator, not the tool. Both tools' working steps, including the output that shaped the artist lists, were not kept, and their final code was a rewrite of what they ran; run A's rewrite gave the same numbers, but a rewrite could hide a step that mattered. Run B also used newer libraries (Python 3.15, scikit-learn 1.9.1), though its figures repeated under Python 3.13. Each tool was run only once, so the runs show how much two AI analysts disagree, not how much one tool varies between runs. In the authors' setup, a Windows checkout let Git convert the CSV's line endings, breaking its SHA-256 check; a `.gitattributes` rule now stores data files byte-for-byte. To repeat the AI analysis a reader needs the tool, model, date, settings, input hash, exact prompts (Appendix A) and full logs, all of which are supplied; even then, the answers may differ.

VI. Conclusion

Fewer than half the rows in this widely used prompt dataset are prompts, three in four repeat an earlier row, and most test rows also appear in training, so it must be cleaned and re-split before use. Given the same data and questions, the AI tools matched the scripted pipeline on every quantity with one obvious definition and diverged on every quantity that needed a judgement, in one case enough to change the conclusion. Anyone using an AI tool for analysis should ask it to state each definition before it computes anything, keep the full log, and re-run the analysis with independent code before reporting it.

Use of AI Tools

The AI-assisted analysis (Part 1a) used the two tools described in Section III-C; the prompts are in Appendix A and the full logs are supplementary files. The scripted pipeline (Part 1b) and the first draft of this paper's structure were written with help from Claude Opus 5.5 (Anthropic), through Claude Code. Claude also scored both runs against the pipeline's results and re-ran checked that scoring by hand. The authors reviewed and ran all code, checked every reported number against the pipeline output and the logs, and are responsible for the final content.

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Appendix A
Prompts Given to the AI Tool

Each prompt was pasted exactly as below, in order, in one new chat. If the tool had stopped, the only allowed reply was "Please continue." The full run logs, the tool's code and the scoring sheet are supplied as supplementary files (`ai-analysis/runs/`).

P0. I have uploaded a dataset of text-to-image prompts collected from the Midjourney Discord server. Analyse this dataset and report your main findings.

P1. How clean is this dataset? Report how many rows are empty, duplicated, or are not real prompts, and whether rows in the validation and test splits also appear in the train split. Explain exactly how you decided what counts as "not a real prompt".

P2. Excluding rows that are not real prompts and removing duplicates, describe what a typical prompt looks like: its length in words, how often Midjourney parameters such as `--ar`, `--no`, `--w`/`--h`, `--stop` and `--seed` are used, and the most common aspect ratios.

P3. What share of prompts reference a named artist or "in the style of" someone? Which artists are named most often? What share use generic quality phrases such as "trending on artstation", "8k" or "octane render"? State the rule you used to detect each.

P4. Identify the main themes or topics in the prompts using a topic model. Report the topics with their top words and the share of prompts in each. How stable are these topics if the analysis is re-run?

P5. Test two hypotheses and report the test, test statistic, p-value and an effect size for each: (H1) prompts that use at least one parameter are longer than prompts that use none; (H2) prompts that name an artist are more likely to use a quality phrase.

P6. Give me the complete code you ran for all of the above, in one block, so that I can reproduce your results.

Appendix B
Reproducibility Details

Data. `succinctly/midjourney-prompts`, Parquet files downloaded from Hugging Face on 6 Nov. 2026. SHA-256: train `c13d7d1e...0a62b`, validation `553a0a02...9c5539`, test `57f8a944...490eb0`. The combined CSV given to the AI tool has SHA-256 `582e6d09...934806` with LF line endings.

Environment. Python 3.8.10, NumPy 1.24.4, pandas 2.0.3, SciPy 1.10.1, scikit-learn 1.3.2, Matplotlib 3.7.5, PyArrow 14.0.2; random seed 9200.

Commands. `python3 analysis/fetch_data.py` downloads and checks the files; `python3 analysis/pipeline.py` writes every number, table and figure; `latexmk -pdf` builds this paper.